

Our land. Our energy future.

Investigating energy resilience through electricity, seasonal resources, land constraints, industrial systems and open modelling.

THE RESEARCH QUESTION

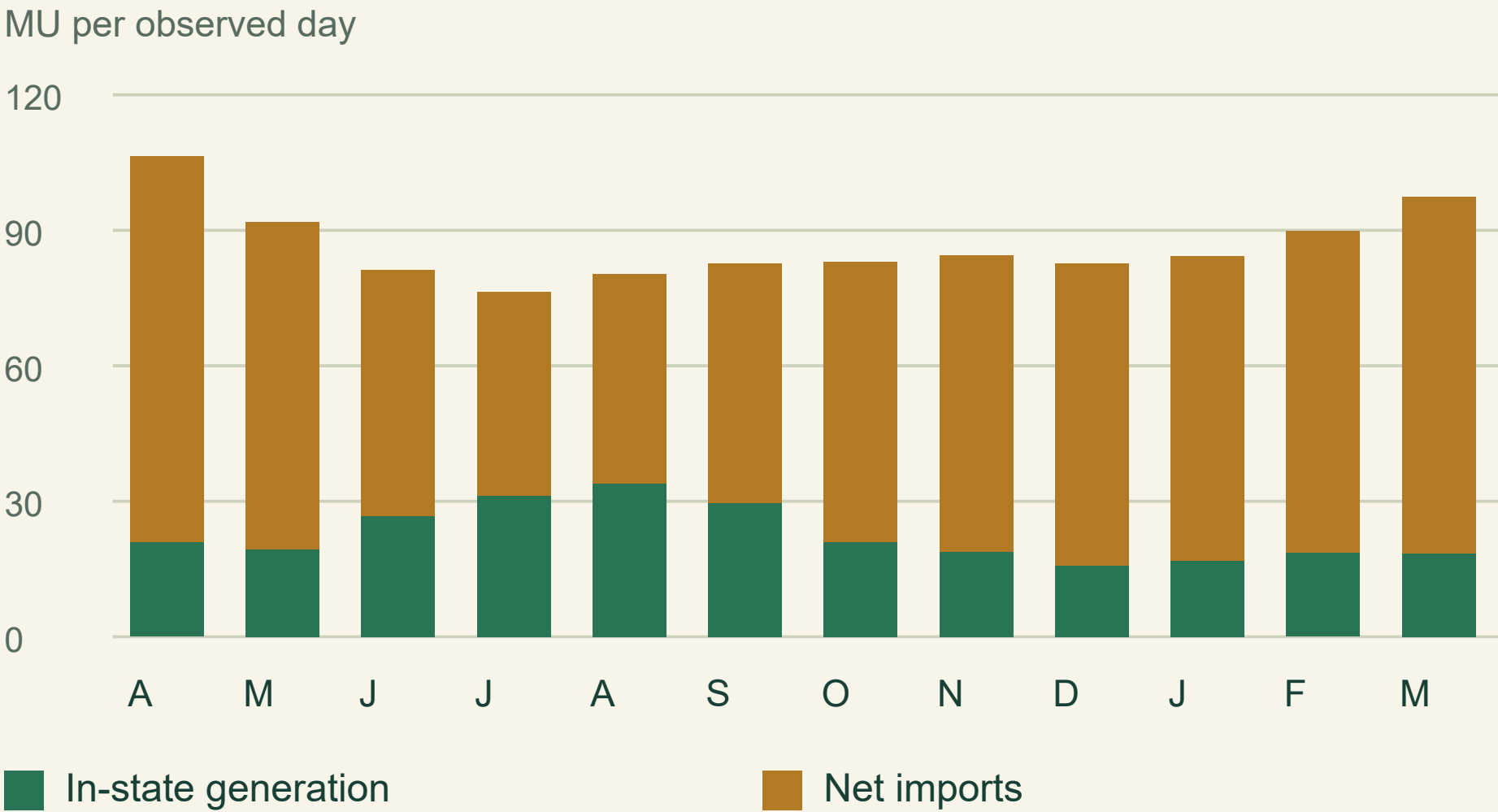
How can Kerala meet future demand reliably while respecting its land and water?

Early experiments show why timing and transfer capability matter alongside new generation. The current evidence supports comparison, not a final 2040 capacity plan.

01 / OBSERVED ELECTRICITY

A connected, seasonal system

73.8% of recorded consumption came from net imports.

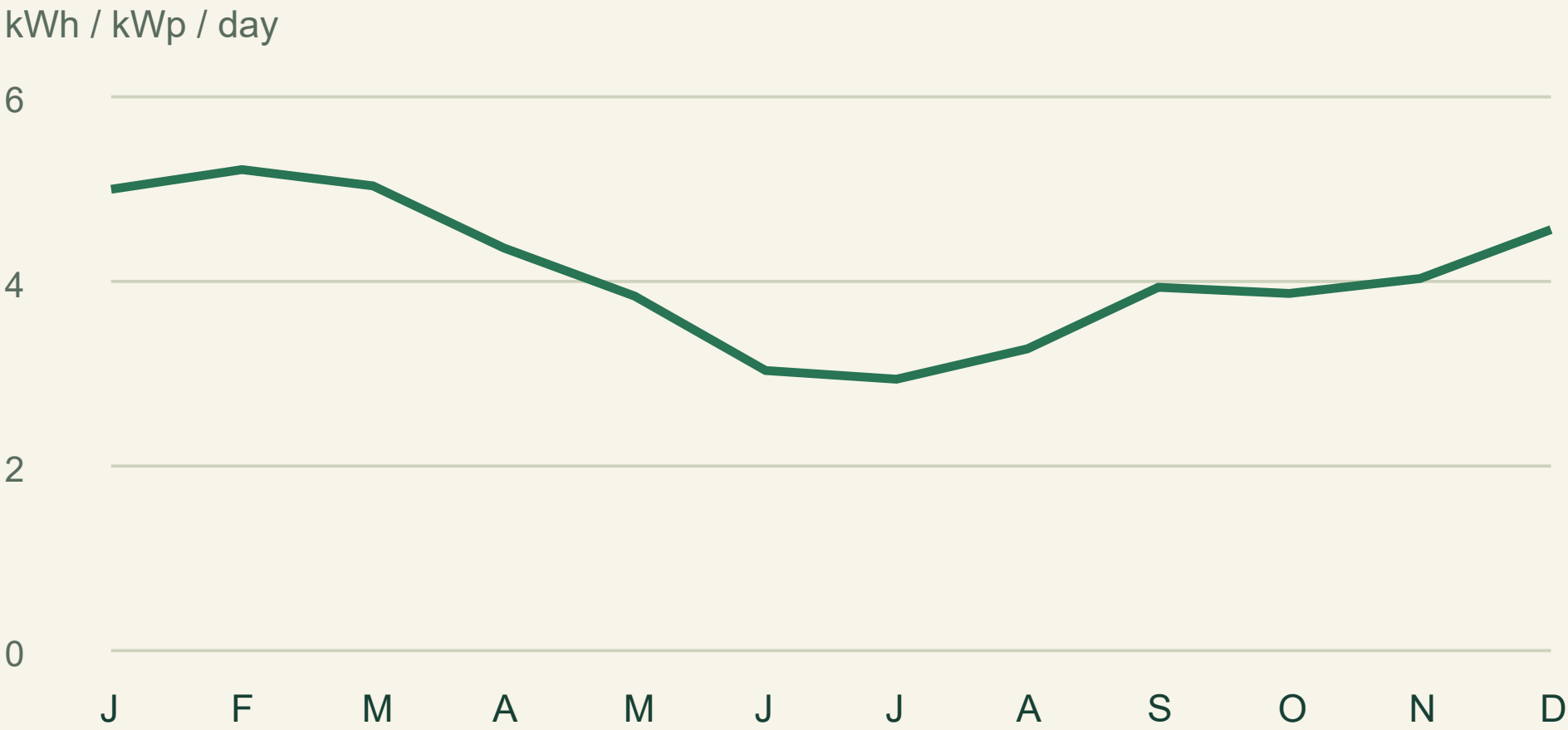


SLDC FY2024-25: 354 of 365 daily reports. The 11 missing days are not filled. Bars show available-day means, not complete monthly totals.

02 / RESOURCE & LAND

Solar follows the seasons

Monthly median source-grid PV output



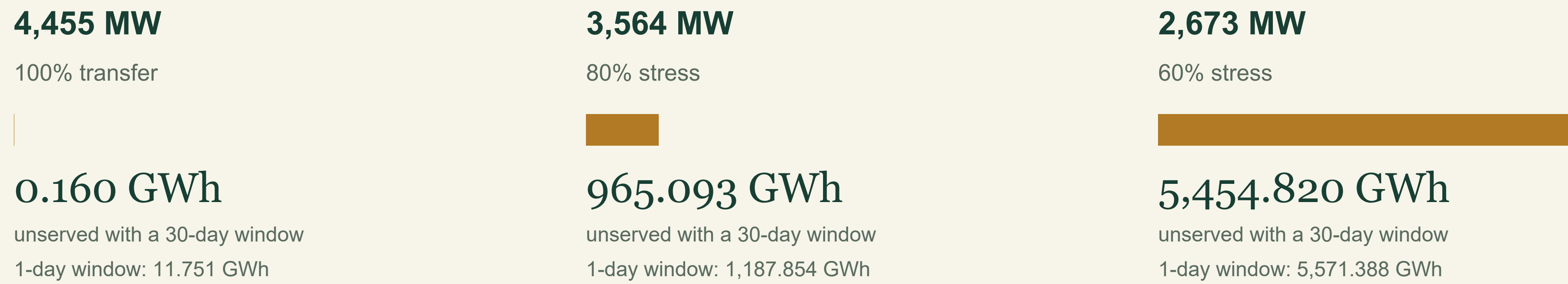
Resource is not permission to build.

Global Solar Atlas 2, 1999-2018 climatology. Wind, terrain, ecology and grid access must also be checked. Eligible land and project capacity are unresolved.

03 / NEW HYDRO-TIMING EXPERIMENTS

Timing helps. Transfer limits still matter.

48 full-year PyPSA cases preserve annual hydro energy while testing synthetic 1, 3, 15 and 30-day timing windows. Window length is not reservoir storage duration.



Reference FY2030-31 demand, full hydro power availability, reference renewable envelope, low battery cost and KSEBL import-price proxy. Synthetic timing bounds; no reservoir water balance, cascades or validated capacity plan.

METHOD

Preserve source records and gaps. Characterise resources. Test constrained dispatch and investment. Check numerical equivalence. Keep assumptions traceable.

WHAT STILL NEEDS WORK

Measured hourly demand; reservoir and cascade operation; outages; landed prices; statutory siting; project finance and grid feasibility.